

NAME: CLASS:.....

DP014
Physics 1
Semester I
Session 2024/2025
2 hours

DP014
Fizik 1
Semester I
Sesi 2024/2025
2 jam



UNIT FIZIK
KOLEJ MATRIKULASI SELANGOR

PRA-PEPERIKSAAN SEMESTER PROGRAM MATRIKULASI 1
MATRICULATION PROGRAMME PRE-EXAMINATION 1

JANGAN BUKA KERTAS SOALAN INI SEHINGGA DIBERITAHU.
DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO

NO SOALAN	MARKAH
1	/8
2	/12
3	/9
4	/12
5	/9
6	/12
7	/18
TOTAL	/80

Kertas soalan ini mengandungi 8 halaman bercetak.
This question paper consists of 8 printed pages.

Answer **all** questions.

1. (a) Convert 3.5 MHz h^{-1} into S.I unit.

[2 marks]

(b)

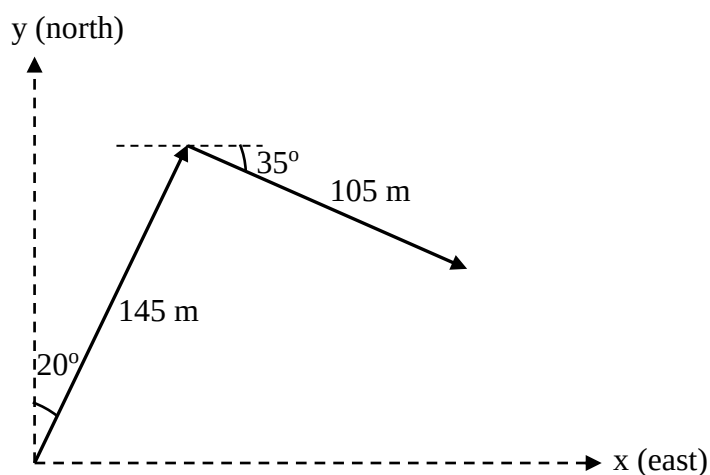


FIGURE 1

FIGURE 1 shows a student running 145 m in a direction 20° east of north and then 105 m in a direction 35° south of east. Determine the resultant vector for these two displacements.

[6 marks]

2. (a) A train initially at rest, accelerates uniformly until its speed reaches 8 m s^{-1} in 25 s. For the next 200 s, the train continues its journey with constant speed, before it slows down uniformly and comes to a complete stop in 20 s.
- Sketch a labelled graph of speed versus time for the whole journey.
 - Calculate the deceleration of the train.
 - Determine the total distance travelled by the train.

[7 marks]

- (b) A girl is throwing a coin into the air from a 8 m building with a velocity of 2.5 m s^{-1} .
- Find the maximum height reach by the coin from the ground.
 - Find the velocity of the coin just before it reaches the ground.

[5 marks]

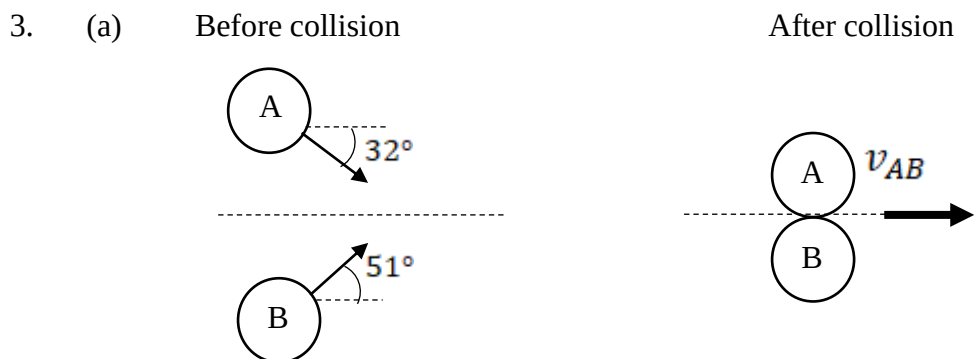


FIGURE 2

FIGURE 2 shows a 0.2 kg ball A moving at 0.6 m s^{-1} makes an angle of 32° below the horizontal while 0.3 kg ball B moving with 0.5 m s^{-1} makes an angle of 51° above the horizontal. Determine the speed of the balls after the collision.

[3 marks]

- (b) A rubber ball of mass 20 g moving at a velocity of 30 m s^{-1} hits a wall and bounces back with the same velocity. Calculate
- the change of momentum.
 - the impulse on the wall.
 - the average force acting on the rubber ball while in contact with the wall for 0.6 s .

[6 marks]

4. (a)

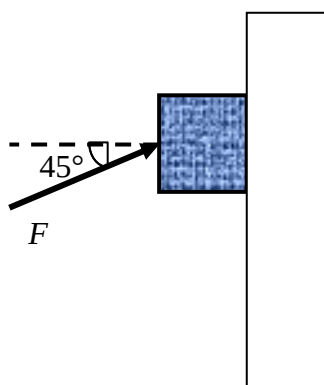


FIGURE 3

FIGURE 3 shows a 0.4 kg block being pushed against a rough vertical wall by a force F at an angle 45° with respect to the horizontal. The block move upwards with acceleration 2 m s^{-2} .

- Sketch a free body diagram of all the forces acting on the block.
- If the coefficient of static friction, $\mu_s = 0.20$, what is the magnitude of F ?

[6 marks]

(b)

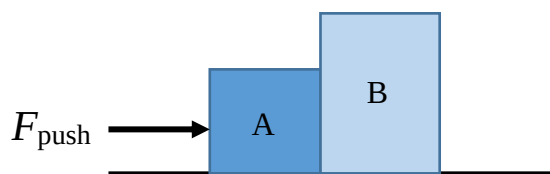


FIGURE 4

Two boxes, A of mass 3.5 kg and B of mass 4.0 kg, are side by side and in contact with each another. They are pushed along a smooth floor under the action of a constant force, $F = 20 \text{ N}$ applied to box A as shown in **FIGURE 4**.

- i. Sketch a free-body diagram for each box.
- ii. Calculate the acceleration of the boxes.
- iii. Determine the magnitude of contact force between box A and box B.

[6 marks]

5. (a) The object takes 25 s to go round the circular track of radius 0.7 m, find
- i. the frequency.
 - ii. the angular velocity.
 - iii. the speed.

[4 marks]

- (b) Two ice skaters, each of mass 60 kg hold hands and spin in a circle with a period of 4.3 s. The distance between them is 1.44 m.

- i. Calculate the angular velocity
- ii. Calculate the centripetal acceleration
- iii. Calculate the centripetal force that acts on each hand

[9 marks]

6. (a) A wheel turns about its axis of rotation at uniform angular acceleration from rest to reach an angular velocity of 30 rad s^{-1} in 20 s. Calculate
- i. the angular acceleration.
 - ii. its tangential acceleration at a point 12 cm from the axis of rotation.
 - iii. its angular displacement after 4 s.

[4 marks]

(b)

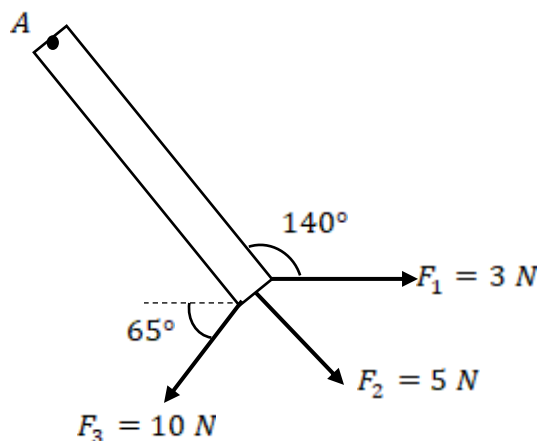
**FIGURE 5**

FIGURE 5 shows a pipe with a length of 15 cm is fixed at point A. Find

- the torque produced by F_1 , F_2 and F_3 .
- the total torque and its direction at point A.

[8 marks]

7. (a) A perfectly insulated aluminium rod has length 50 cm and cross-sectional area 3.0 cm^2 . The temperatures of the surfaces are 150°C and 50°C . (Thermal conductivity of aluminium is $210\text{ Wm}^{-1}\text{ K}^{-1}$)
- Sketch a labelled graph of temperature against distance.
 - Calculate the temperature gradient along the rod.
 - Calculate the rate of heat flow in the rod.

[6 marks]

- (b) A balloon is filled with helium at 25°C . The mass of a helium atom is $6.65 \times 10^{-27}\text{ kg}$. Calculate the root mean square speed of the helium atoms.

[2 marks]

- (c) A 15 liter gas cylinder contains helium gas with pressure $1.01 \times 10^5\text{ Pa}$ at 25°C . When heated, the gas undergoes an isochoric process.
- If 500 J of heat is added, calculate the change in the internal energy of the gas.
 - Sketch the graph of pressure versus volume for the isochoric process.

[4 marks]

(d)

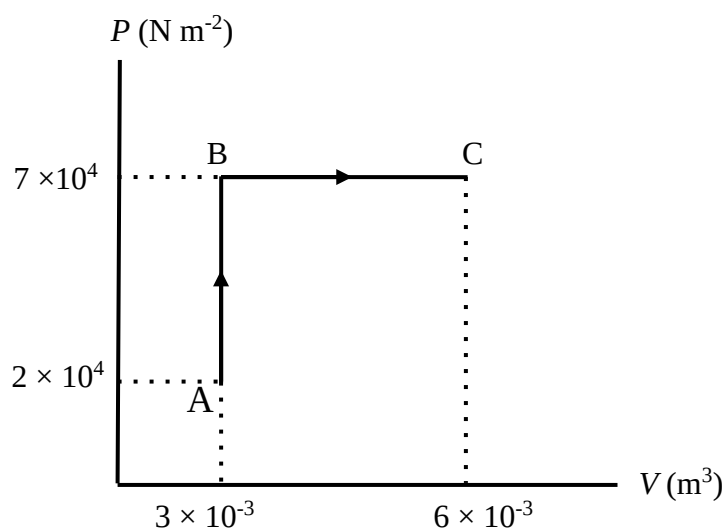


FIGURE 6

FIGURE 6 shows a series of thermodynamic processes ABC. During the process AB and process BC, 120 J of heat and 500 J of heat are added respectively. In process BC, 250 J of work is done on the system.

- Name the process AB and process BC.
- Calculate the change in the internal energy in the process AB and BC.

[6 marks]

END OF QUESTION PAPER

KERTAS SOALAN TAMAT